

maximum sunlight, the prices just go through the roof.

Thin-film solar cells, often dubbed as the next generation of solar technology, are another option over older silicon models. These are made by layering a thin solar absorbent material over a substrate such as glass or plastic, which can be flexible. The materials being used in the present thin films are, however, a concern. Cadmium in cadmium-telluride (CdTe) is highly toxic, making handling and placement of solar cells a major concern.

The poisonous potential.

Solar, an American photovoltaic manufacturer of rigid thin-film modules, has announced an efficiency of 22.1 per cent using experimental cells made from CdTe. It is a thin-film technology, requiring less material to produce a comparable amount of electricity than conventional silicon technology. It is also poisonous to humans. The poisonous material has proven to be highly efficient; in labs, at least. So it would be a boon if we could work around these materials.

Manufacturing process for CdTe solar cells is also simpler, resulting in lower costs for electricity production—purely theoretical, of course. Trade-offs seem to be pretty high there.

Some low-cost solutions. Tara Dhakal, interim director, Centre for Autonomous Solar Power, and assistant professor, department of electrical and computer engineering, Binghamton University, UK, is working with pyrite and copper-zinc-tin-sulfide (CZTS) as these are non-toxic and available in abundance. He has explained the details in a paper earlier this year. The paper reads, "CZTS costs about 0.005 cents per watt, and pyrite costs a mere 0.00002 cents per watt." The biggest advantage is that, thickness of these chips can go as low as a thousandth of a millimetre.

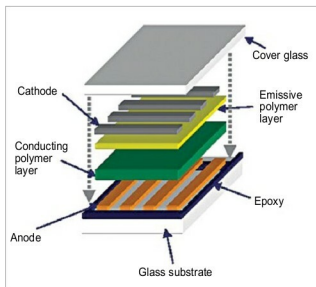


Fig. 2: OLED structure

In a study earlier this year, researchers at Stanford University, USA, and University of California at Berkeley, USA, estimated that, by 2050, 45 per cent of electricity in America could be dependent on solar energy. Fool's gold seems to be taking over liquid gold, at least in the western hemisphere.

Organics in lighting

Organics are coming in use in lighting as well. Moving on from regular LEDs, we now have OLEDs with high luminescence. Some of the best displays in 2016 have been various variants of OLEDs. AMOLED and super-AMOLED are premium lighting devices with amazing clarity. The story behind OLED, however, deals with a lot of chemistry.

Pyrimidine is an aromatic heterocyclic organic compound similar to pyridine that is marking significant changes in OLEDs. Pyrimidine-containing thermally-activated delayed-fluorescence (TADF) emitters are being used to bring out improvements in efficiency. OLEDs based on blue pyrimidine based TADF emitters have exhibited high external quantum efficiency of 20.8 per cent and power efficiencies of 31.5lm/W.

Researchers from National Tsing-Hua University, Taiwan, are working with two benzoylpyridine-carbazole

based fluorescent materials, DCBPY and DTCBPY. These molecules have shown smaller ΔE_{ST} of 0.03 and 0.04eV and transient photoluminescence (PL) characteristics. These properties relate to TADF materials. These are also photoluminescent, with quantum yields of 14 to 36 per cent in cyclohexane. Values rise rapidly from 88 to 91.4 per cent in thinner-film variants. OLEDs using DCBPY and DTCBPY for doping emit blue and green lights, creating a possible solution for a stable blue light source in OLEDs.

With proper application, you might very well be living in Tony Stark's mansion with windows working as screens to display notifications or other tasks!

3D printing for fun

Poly(lactic acid) (PLA) is a popular biodegradable thermoplastic derived from renewable resources. It is also more environmentally friendly among other plastic materials. PLA is harder than acrylonitrile-butadiene-styrene (ABS) and melts at 180°C to 220°C. The most popular thermoplastic, however, is ABS. It is very durable, slightly flexible and lightweight, and can be easily extruded.

However, the catch with ABS is the higher temperature requirement of about 210°C to 250°C. Nylon 618 and 645 are also in the running due to their ease of use and lower temperatures to work with. Designed to simulate ABS plastic, tough resin from Formlabs provides comparable tensile strength and modulus required for functional prototyping.

Aesthetically-pleasing applications

Thermoplastic elastomers (TPEs) like ninjaflex and filaflex provide options for flexible 3D printing. However, printed cells have shown to be vulnerable to moisture, which could lead to contamination if a cell